

## MAP — $\sigma$ -essential descent

*Read this FIRST. 30 seconds to recover the shape. One task, three exits, one cheap next action. Last synced 2026-06-22 (rewritten around the single task). If this drifts from the working notes, the working notes win — but fix this.*

### The target (the whole thing, one sentence)

Is there a concrete  $\sigma$ -complete OML carrying a  $\sigma$ -additive contextual state witnessed by no finite sub-OML?

**Equivalently (the sharp form):** a concrete  $\sigma$ -complete OML  $L$  with a *restriction-gap*  $\{s|_F : s \in S_{\text{df}}^\sigma\} \subsetneq S_{\text{df}}(F)$  for some finite  $F$  — i.e. global  $\sigma$ -additive dispersion-free states fail to restrict onto the local df-state hull. (=  $\sigma$ -additive lifting failure.)

**STATUS: OPEN, PARKED.** Not solved, not proved impossible.

### You are here

Both sides of the duality are determined at the representation level (clauses (i)+(ii), settled + inhabited by  $\prod_n MO_2$ ). The entire weight sits on **clause (iii) = the measure descent = rem:dw residue = the one task below**. The three “doors” of the prior map are *not* three routes — they are the three possible **EXITS** of one question. The predicate fork (P-weak vs P-strong) is **resolved** (2026-06-22) and collapsed *into* that one task.

### The one thing to do (everything reduces to this)

Construct — or refute the existence of — a binding global  $\sigma$ -additive 2-valued point over a concrete  $\sigma$ -complete non-Boolean OML.

This is **Door 1. It is the only way through.** Door 2 (a candidate object) is the *same construction* (an object is a witness only if it carries the point); Door 3 can’t even be *stated* until the object exists. The three exits:

| If Door 1 resolves as...                            | ... the exit is               | name   |
|-----------------------------------------------------|-------------------------------|--------|
| the point EXISTS (construct it)                     | a <b>witness</b>              | Exit A |
| exists in one ZFC model, fails in another (forcing) | an <b>independence</b> result | Door 3 |
| the point provably CANNOT exist                     | an <b>impossibility</b>       | Exit B |

**You do not choose an exit. You work Door 1 and find out which one it has.**

### How Door 1 must be attacked (the tool is forced)

**By-hand / compositional construction is CLOSED** — every elementary assembly hits a horn (the graveyard in “Settled” below). The lever must be GLOBAL and NON-CONSTRUCTIVE: any pairwise/finite-assembly spec of  $A \vee B$  across non-orthogonal pairs collapses to a closed horn, so the point cannot be *built* locally — it must be *summoned* globally.

⇒ **The only tool left is FORCING** (or an inner-model argument). The missing object and the independence proof are plausibly the *same* forcing construction. **Where it bites:** only at the UNCOUNTABLE-BLOCK /  $S_{\text{df}}^\sigma$ -density level — on  $\mathbb{N}$  the obstruction is a clean ZFC contradiction ( $\bigcap A_k = \emptyset$ ), no independence there.

**Honest cost:** months-long, set-theory-heavy (iterated forcing, plausibly a measurable cardinal or its failure); real chance the answer is “the object can’t be set up coherently” rather than a clean theorem.

### The actual next action (cheap, before the months-long commitment)

**FORCING SCOUT** — do NOT start the heavy construction cold. Bounded effort first:

- (1) **Pin the precise set-theoretic sentence** whose independence you’d prove (“a binding  $\sigma$ -point exists over [object schema]”). Check it isn’t already decided.
- (2) **Survey what forcing / large-cardinal tech could apply** to  $\sigma$ -point existence over non-distributive structures (RVM cardinals, Boolean-valued models, Blecher–Weaver *no-RVM* ⇒  $\sigma$ -states restrict normally, the  $\text{add}(\mathcal{N})$  lead — but  $\text{add}(\mathcal{N}) \Rightarrow$  independence was already fenced as NOT earned).

Outcome: a **FOOTHOLD** (a forcing notion that plausibly acts) or **CONFIRMATION THE WALL IS LOAD-BEARING**. Both worth the low cost; neither commits the months. **Recommended before opening a forcing branch.**

**The persist-vs-pivot call is yours** (problem-selection = human): commit to the forcing programme now, run the scout first, or confirm the park. No rush.

### The standing guardrails (violating these = the failure mode)

- **rem:dw is STABLE.**  $\sigma$ -essential cell EMPTY for Polish/regular-representable OMLs (Derr–Williamson). Residue = non-representable case =  $\sigma$ -LS wall. Restate, never reopen.
- **“Couldn’t build it” ≠ “needs choice” ≠ “independent of ZFC.”** Three claims, increasing strength. We have the first (proved), suspect the second (conjecture), have NOT shown the third (= Door 3, the open work). Don’t slide up the chain without doing the math. (Nor couldn’t-refute as NOT earned.)
- **CARRYING vs SELECTING** (the 2026-06-22 Door-1 discriminator): the difficulty is NOT that blocks can’t hold the mass — states are plentiful ( $S_{\text{df}}$  compact,  $\prod_n MO_2$  state-rich; “horizontal-sum-destroys-states” is a KILLED premise). It is that **SELECTING** one globally-coherent  $\sigma$ -additive point across all incompatible block-pairs at once is the non-compact inverse-limit / free-ultrafilter existence step. Door 1 is a *selecting* problem.
- **Exit-B inclination held OPEN, not a verdict.** ~23 reversals on record.
- **Red-flag rule:** on ANY positive result, first ask *where did the smuggle enter?* — inner-regularity / a  $\pi$ -system / the  $P(A) \leftrightarrow S_{\text{df}}^\sigma$  switch / a finitary obstruction masquerading as  $\sigma$ -essential.
- All novel hand-work flagged **[HAND]**; user verifies. Survey gets NO unproven claim.

— end of the action map (*p.1 = the whole shape: one task, three exits, the forcing scout*). Page 2 is the reference annex: settled facts, file pointers, persist-vs-pivot. —

## Settled — do NOT re-walk (most of the ~23 reversals, now closed)

- **Deliverable = the restriction-gap** (above). Non-Borelness of  $S_{\text{df}}^\sigma$  is only its descriptive shadow. Off-hull is ALWAYS finitely detected (Hahn–Banach in  $\mathbb{R}^L$ ).
- **Duality determined at (i)+(ii)** = Derr–Williamson  $\sigma$ -Dynkin systems, inhabited by  $\prod_n MO_2$ . A definitional pin, not a theorem; all weight on (iii).
- **Predicate fork RESOLVED** (P-weak  $\Rightarrow$  P-strong; co-extensive  $\Leftrightarrow$   $\sigma$ -point-realization = Door 1; agree on all segregated/Polish objects; diverge only on the missing witness; use P-strong if a witness ever appears). A restatement of Door 1, not a new door.
- **Countable-generation  $\Rightarrow$  lifting: PARKED** (prior-art OPEN; tribe-vs-points structural; cardinality-control axis DISSOLVED by Burešová–Pták). Witness, if any, uncountably generated.
- **Order-determination does NOT help** (Paper-I Stone-descent transfer):  $\delta_x$  are order-determining for free yet lifting can fail — it’s the settled (i)+(ii) leg, not the descent leg. The OML lacks the Stone-DUALITY leg = Door 1.
- **All carrier constructions dead:** faithful  $\sigma$ -tribe (Floor); 5 binding modes; band family (S8 dichotomy, Lean-certified); loops/Greechie/KS (finite-character). *This is why the lever must be global/non-compositional  $\rightarrow$  forcing.*
- **Strategy D KILLED** (prior-art, Gaifman 1964). Separate from the descent arc.

## Reference annex — the doorway map (which file answers which question)

| If you’re asking...                          | Go to                                                   |
|----------------------------------------------|---------------------------------------------------------|
| “What’s the shape / what do I do?”           | <b>this file</b>                                        |
| “Live record of the park + forward pointer?” | <code>sigma_duality_targets.md</code> §9, §9a           |
| “The two sides of the duality?”              | <code>sigma_duality_targets.md</code> §1–5              |
| “The predicate fork resolution?”             | <code>sigma_duality_targets.md</code> §6 RESOLVED block |
| “Has new input arrived / what un-parks?”     | <code>NEXT_SESSION_sigma_essential.md</code>            |
| “Why is each construction route dead?”       | <code>sigma_essential_construction_attempt.md</code>    |
| “Is it open / prior-art?”                    | <code>sigma_essential_prior_art_verdict.md</code>       |
| “The authoritative math write-up”            | <code>oml_onboarding.{tex,pdf}</code> (survey)          |
| “Work it by hand”                            | <code>problemset_oml_descent.{tex,pdf}</code>           |

## The non-math fork (if the weight is “right problem?” not “what do I do?”)

Three moves, all YOUR call (problem-selection = human):

- **(a) Commit to the forcing programme now** — open a branch, start the heavy set-theory work on Door 1. Large, real; may end in “the park is deeper.”
- **(b) Run the forcing scout first** (p.1) — bounded cost, finds a foothold or confirms the wall is load-bearing. *Recommended.*
- **(c) Confirm the park and step away.** Honest; the method named the residue.

This is thesis-advisor territory (persist vs pivot), not something to settle by guessing.